Introduction

The 4-color problem can be restated in terms of graphs. Choose a vertex-point in the interior of each country, and join two vertex-points by an edge-curve whenever the two countries share a border. This this way, we obtain a plane-graph (hence a planar graph) which is called the *dual graph* of the map.



Chromatic number I

- Let $G = (V, E)$ be a graph. A *vertex-coloring* of G is assignment of a color to each of the vertices of G in such a way that adjacent vertices are assigned different colors. If the colors are chosen from a set of k colors, then the vertex-coloring is called a *k-vertex-coloring*, abbreviated *k-coloring*, whether or not all k colors are used. If G has a k-coloring, then G is said to be *k-colorable*. The smallest k, such that G is k-colorable, is called the *chromatic number* of G, denoted by $\chi(G)$.
- A *null graph* is defined to be a graph without any edges. A null graph of order n is denoted by N_n.

Theorem

Let G be a graph of order $n \geq 1$. Then

$$1 \leq \chi(G) \leq n.$$

Moreover, $\chi(G) = n$ if and only if G is a complete graph, and $\chi(G) = 1$ if and only if G is a null graph.

Chromatic number II

Corollary

Let G be a graph and let H be a subgraph of G. Then $\chi(G) \geq \chi(H)$. If G has a subgraph equal to a complete graph K_p of order p, then

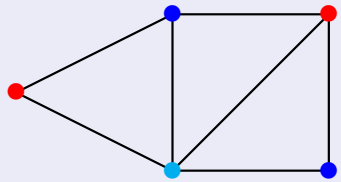
$$\chi(G) \geq p.$$

- Let $G = (V, E)$ be a graph that is k-colored, using the colors $1, 2, \ldots, k$. Let V_i denote the subset of vertices that are assigned the color i, $(i = 1, 2, \ldots, k)$. Then $V_1, V_2, \ldots, V_k$ is a partition of V, called a *color partition* for G. Moreover, the induced subgraphs G_{V_1}, G_{V_2}, ..., G_{V_k} are null graphs. Another way to describe the chromatic number of G is that $\chi(G)$ *is the smallest integer k such that the vertices of G can be partitioned into k sets with each set inducing a null graph*.



Chromatic number III

Example



Corollary

Let G be a graph of order n and let q be the largest order of an induced subgraph of G equal to a null graph N_q. Then

$$\chi(G) \geq \left\lceil \frac{n}{q} \right\rceil .$$

Theorem

Let G be a graph with at least one edge. Then $\chi(G) = 2$ if and only if G is bipartite.

Chromatic number IV

Example

A scheduling problem – Many scheduling problems can be formulated as problems that ask for the chromatic number (but often will settle for a number not much larger than the chromatic number) of a graph. The basic idea is that we associate a graph with a scheduling problem whose vertices are the "tasks" to be scheduled, putting an edge between two tasks whenever they conflict, and hence cannot be scheduled at the same time. A color partition for G furnishes a schedule without any conflicts. The chromatic number of the graph thus equals the smallest number of time slots in a schedule with no conflicts.

Greedy algorithm for vertex-coloring

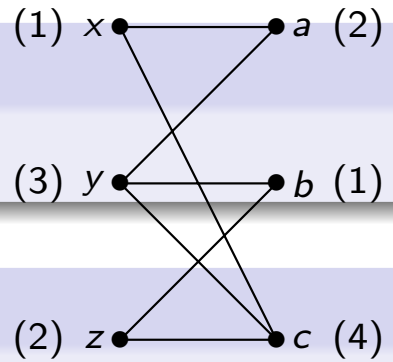
Let G be a graph in which the vertices have been listed in some order $x_1, x_2, \ldots, x_n$.

1. Assign the color 1 to vertex x_1.
2. For each $i = 2, 3, \ldots, n$, let p be the smallest color such that none of the vertices $x_1, x_2, \ldots, x_{i-1}$ which are adjacent to x_i is colored p, and assign the color p to x_i.

Chromatic number V

Example

Vertices are listed as x, a, b, y, z, c.



Theorem

Let G be a graph for which the maximum degree of a vertex is Δ. Then the greedy algorithm produces a $(\Delta + 1)$-coloring of the vertices of G, and hence

$$\chi(G) \leq \Delta + 1.$$

Chromatic number VI

- There are two classes of graphs, the complete graphs K_n, for which $\Delta = n - 1$ and $\chi(G) = n$, and the graphs C_n of odd order n whose edges are arranged in a cycle (of odd length), for which $\Delta = 2$ and $\chi(G) = 3$.

Theorem

Let G be a connected graph for which the maximum degree of a vertex is Δ. If G is neither a complete graph K_n nor an odd cycle graph C_n, then $\chi(G) \leq \Delta$.

- For each nonnegative integer k, the number of k-colorings of the vertices of a graph G is denoted by $P_G(k)$. If $\chi(G) > k$, then $P_G(k) = 0$. For example, for a complete graph, we have

$$p_{K_n}(k) = k(k-1)\cdots(k-(n-1)) = [k]_n,$$

since each vertex must be a different color. For a null graph, we have

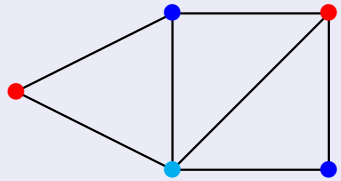
$$p_{N_n}(k) = k^n,$$



Chromatic number VII

since we can arbitrarily assign colors to each of the vertices.

Example



$P_G(k) = k(k-1)(k-2) \times (k-2) \times (k-2) = k(k-1)(k-2)^3$.

Theorem

Let T be a tree of order n, Then

$$p_T(k) = k(k-1)^{n-1}.$$

- Indeed, this is no accident and is a general phenomenon that $p_G(k)$ is always a polynomial function of k. As a result, $p_G(k)$ is called the *chromatic polynomial* of the graph G.



Chromatic number VIII

- The chromatic number of G is the smallest nonnegative integer that is not a root of the chromatic polynomial.
- Let G be a graph and let $\alpha = \{x, y\}$ be an edge of G. We now denote the graph obtained from G by deleting the edge α by $G_{\ominus\alpha}$. We also denote the graph obtained from G by identifying x and y by $G_{\otimes\alpha}$. We say that $G_{\otimes\alpha}$ is obtained from G by *contracting* the edge α. We have the relation

$$p_G(k) = p_{G_{\ominus\alpha}}(k) - p_{G_{\otimes\alpha}}(k).$$

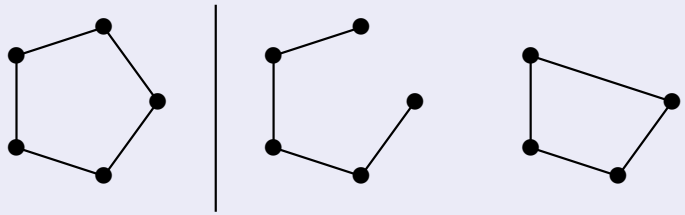
Repeated use of deletion and contraction gives an algorithm for determining $p_G(k)$. We consider objects $(\pm, H)$, where H is a graph. We call such an object a *signed graph*, a graph with either a plus sign $+$ or minus sign - associated with it.



Chromatic number IX

Example

$$\begin{aligned} P_G(k) =& k(k-1)^4 - k(k-1)^3 \\ &+ k(k-1)(k-2) \\ =& k(k-1)(k-2)(k^2-2k+2). \end{aligned}$$



Algorithm for computing the chromatic polynomial of a graph

Let $G = (V, E)$ be a graph.

1. Put $\mathcal{G} = \{(+, G)\}$.
2. While there exists a signed graph in $\mathcal{G}$ that is not a null graph, do the following,
   1. Choose a nonnull signed graph (ϵ, H) in $\mathcal{G}$ and an edge α of H.
   2. Remove (ϵ, H) from $\mathcal{G}$ and put in the two signed graphs $(\epsilon, H_{\ominus\alpha})$ and $(-\epsilon, H_{\otimes\alpha})$.
3. Put $p_G(k) = \sum \epsilon k^p$, where the summation extends over all signed graphs (ϵ, H) in $\mathcal{G}$ and p is the order of H.

Chromatic number X

Theorem

Let G be a graph of order $n \geq 1$. Then, the number of k-colorings of G is a polynomial in k of degree equal to n (with leading coefficient equal to 1) and this polynomial-the chromatic polynomial of G-is computed correctly by the preceding algorithm.

- Let $G = (V, E)$ be a connected graph and let U be a subset of the vertices of G. Then U is called an *articulation set* of G, provided that the subgraph G_{V-U} induced by the vertices not in U is disconnected. A connected graph has an articulation set if and only if it is not complete.

Theorem

Let G be a graph and assume that G contains a subgraph H equal to a complete graph K_r. Then the chromatic polynomial of G is divisible by the chromatic polynomial $[k]_r$ of K_r.



Chromatic number XI

Theorem

Let U be an articulation set of G and suppose that the induced subgraph G_U is a complete graph K_r. Let the connected components of G_{V-U} be the induced subgraphs $G_{U_1}, \ldots, G_{U_t}$. For $i = 1, \ldots, t$, let $H_i = G_{U \cup U_i}$ be the subgraph of G induced by $U \cup U_i$. Then

$$p_G(k) = \frac{p_{H_1}(k) \times \cdots \times p_{H_t}(k)}{([k]_r)^{t-1}}.$$

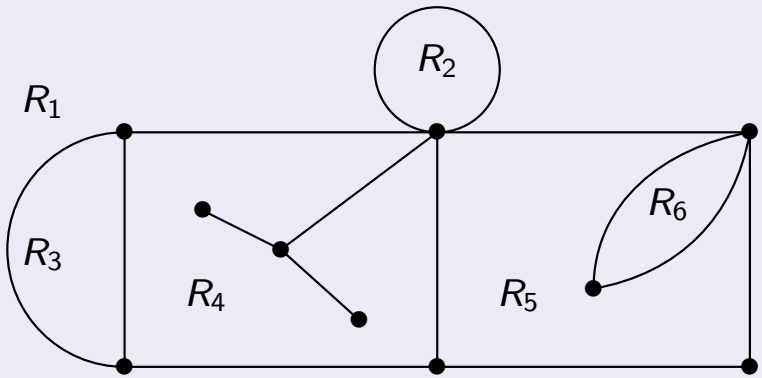
In particular, the chromatic number of G is the largest of the chromatic numbers of $H_1, \ldots, H_t$.



Plane and planar graphs I

- Let $G = (V, E)$ be a planar general graph and let G' be a planar representation of G. The plane graph G' divides the plane into a number of regions that are bounded by one or more of the edge-curves. Exactly one of these regions extends infinitely far.

Example



The number of edge-curves bordering regions is counted in such a way that each edge-curves is counted twice, either because it borders two different regions or because it borders the same region twice.



Plane and planar graphs II

- Let G' be a plane-graph with n vertex-points, e edge-curves, and r regions. Let the number of edge-curves bordering the regions be, respectively,

  $$f_1, f_2, \ldots, f_r.$$

  Then, using the convention established, we have

  $$f_1 + f_2 + \cdots + f_r = 2e.$$

Theorem

Let G be a plane-graph of order n with e edge-curves and assume that G is connected. Then the number r of regions into which G divides the plane satisfies (*Euler's formula*)

$$r = e - n + 2.$$



Plane and planar graphs III

Theorem

Let G be a connected planar graph. Then there is a vertex of G whose degree is at most 5.

- Let $G = (V, E)$ be a nonplanar graph and let $\{x, y\}$ be any edge of G. Let G' be obtained from G by choosing a new vertex z not in V and replacing the edge $\{x, y\}$ with the two edges $\{x, z\}$ and $\{z, y\}$. We say that G' is obtained from G by *subdividing the edge* $\{x, y\}$. If G is not planar, then clearly G' is also not planar. A graph H is called a *subdivision* of a graph G, provided that H can be obtained from G by successively subdividing edges. If H is is a subdivision of G, then we can think of H as obtained from G by inserting several new vertices (possible none) on each of its edge.

Theorem

A graph G is planar if and only if it does not have a subgraph that is a subdivision of a K_5 or of a $K_{3,3}$.



Plane and planar graphs IV

- A graph H is a *contraction* of a graph G provided that H can be obtained from G by successively contracting edges.

Theorem

A graph G is planar if and only if it does not contain a subgraph that contracts to a K_5 or a $K_{3,3}$.



A 5-color theorem I

Theorem

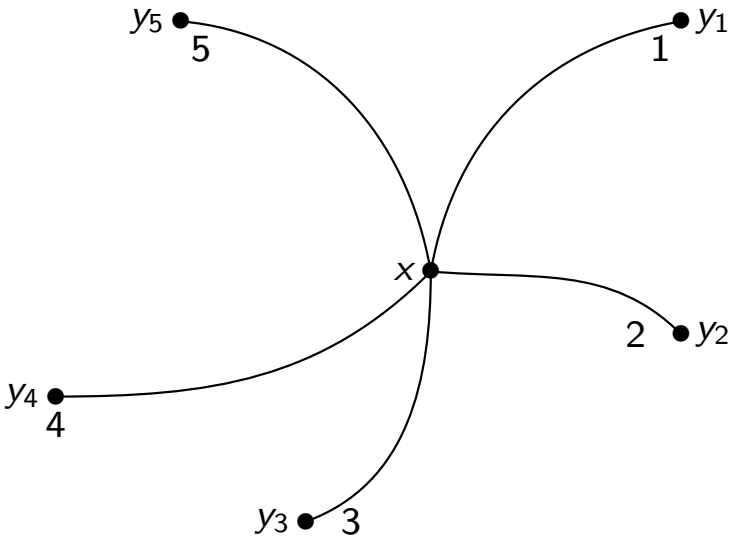
Let there be given a k-coloring of the vertices of a graph $H = (U, F)$. Let two of the colors be red and blue, and let W be the subset of vertices in U that are assigned either the color red or the color blue. Let $H_{r,b}$ be the subgraph of H induced by the vertices in W and let $C_{r,b}$ be a connected component of $H_{r,b}$. Interchanging the colors red and blue assigned to the vertices of $C_{r,b}$, we obtain another k-coloring of H.

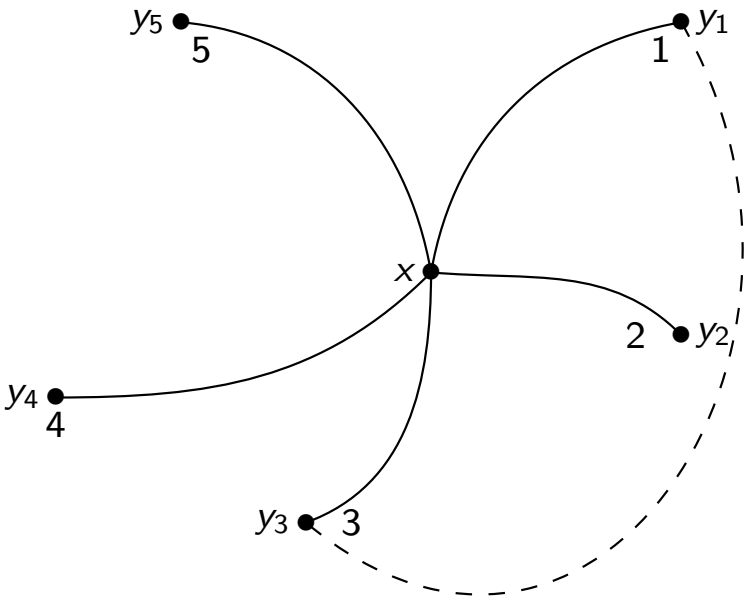
Theorem

The chromatic number of a planar graph is, at most, 5.



A 5-color theorem II





A 5-color theorem III

In 1943, Hadwiger made a conjecture about the chromatic number of graphs, which, except in a few cases, is still unsolved. This conjecture asserts: *A connected graph G whose chromatic number satisfies $\chi(G) \geq p$ can be contracted to a K_p. Equivalently, if G cannot be contracted to a K_p, then $\chi(G) < p$.*

Theorem

Hadwiger's conjecture holds for $p = 5$ if and only if every planar graph has a 4-coloring.

Theorem

Let $p \leq 3$. If G is a connected graph with chromatic number $\chi(G) \geq p$, then G can be contracted to a K_p.



Independence number and clique number I

- Let $G = (V, E)$ be a graph of order n. A set of vertices U of G is called *independent*, provided that no two of its vertices are adjacent, equivalently, provided the subgraph G_U of G induced by the vertices in U is a null graph. The largest number of vertices in an independent set is called the *independent number* of the graph G and is denoted by $\alpha(G)$.
- The independence number is the largest number of vertices that can be colored the same in a vertex-coloring of G. Corollary 13.1.3 can be rephrased as

$$\chi(G) \geq \left\lceil \frac{n}{\alpha(G)} \right\rceil.$$

For a null graph N_n, a complete graph K_n, and a complete bipartite graph $K_{m,n}$, we have

$$\alpha(N_n) = n, \quad \alpha(K_n) = 1, \quad \text{and} \quad \alpha(K_{m,n}) = \max\{m, n\}.$$



Independence number and clique number II

Example

The problem of the 8 queens – Consider an 8-by-8 chessboard and the chess piece known as a *queen*. In chess, a queen can attack any piece that lies in its row or column or in one of the two diagonals containing it. If 9 queens are placed on the board, then necessarily, two lie in the same row and thus can attack one another. Is it possible to place 8 queens on the board so that no queen can attack another?

- Let G be the queens graph of the chessboard. The vertices of G are the squares of the board, with two squares adjacent if and only if a queen placed on one can attack a queen placed on the other. Our question thus asks whether the independent number of the queens graph equals 8. In fact, $\alpha(G) = 8$ and there are 92 different ways to place 8 nonattacking queens on the board.



Independence number and clique number III

- Let $G = (V, E)$ be a graph and let U be an independent set of vertices that is not a subset of any larger independent set. Thus, no two vertices in U are adjacent, and each vertex not in U is adjacent to at least one vertex in U. A set of vertices with the later property is called a dominating set. More precisely, a set W of vertices of G is a *dominating set*, provided that each vertex not in W is adjacent to at least one vertex in W. The smallest number of vertices in a dominating set is called the *dominating number* of G and is denoted by $\mathrm{dom}(G)$.
- For a null graph, complete graph, and complete bipartite graph, we have

$$\mathrm{dom}(N_n) = n, \mathrm{dom}(K_n) = 1, \text{and } \mathrm{dom}(K_{mn}) = 2 \text{ if } m, n \geq 2.$$

Theorem

Let G be a connected graph of order $n \geq 2$. Then

$$\mathrm{dom}(G) \leq \left\lfloor \frac{n}{2} \right\rfloor.$$

Independence number and clique number IV

- A *clique* in a graph G is a subset U of vertices, each pair of which is adjacent, equivalently, the subgraph induced by U is a complete graph. The largest number of vertices in a clique is called the *clique number* of G and is denoted by $\omega(G)$. For a null graph, complete graph, and complete bipartite graph, we have

$$\omega(N_n) = 1, \omega(K_n) = n, \text{and } \omega(K_{mn}) = 2.$$

The notion of a clique of a graph is "complementary" to that of independence in the following sense. Let $\overline{G} = (V, \overline{E})$ be the *complementary graph* of G. Recall that the complementary graph of G has the same set of vertices as G, and two vertices are adjacent in $\overline{G}$ if and only if they are not adjacent in G. It follows from definitions that, for a subset U of V, U is an independent set of G if and only if U is a clique of $\overline{G}$, and U is a clique of G if and only if U is an independent set of $\overline{G}$. In particular, we have

$$\alpha(G) = \omega(\overline{G}) \quad \text{and} \quad \omega(G) = \alpha(\overline{G}).$$



Independence number and clique number V

- The chromatic number and clique number are related by the inequality (cf. Theorem 13.1.2)

$$\chi(G) \geq \omega(G). \tag{13.1}$$

Every bipartite graph G with at least one edge satisfies $\chi(G) = \omega(G) = 2$. A cycle graph C_n of odd order $n > 3$ with n edges arranged in a cycle satisfies $\chi(C_n) = 3 > 2 = \omega(C_n)$.

- A *clique-partition* of a graph G is a partition of its vertices into cliques. The smallest number of cliques in a clique-partition of G is the *clique partition number* of G, denoted by $\theta(G)$. We have

$$\chi(G) = \theta(\overline{G}) \text{and} \quad \theta(G) = \chi(\overline{G}).$$

The inequality "complementary" to that in (13.1) is

$$\theta(G) \geq \alpha(G).$$



Independence number and clique number VI

- A graph G is called χ-*perfect*, provided that $\chi(H) = \omega(H)$ for every induced subgraph H of G. The graph G is θ-*perfect*, provided that $\theta(H) = \alpha(H)$ for every induced subgraph H of G.

Theorem

A graph G is χ-perfect if and only if it is θ-perfect. Equivalently, G is χ-perfect if and only if $\overline{G}$ is χ-perfect.

- As a result of this theorem we now refer to *perfect graphs*, and we show the existence of a large class of perfect graphs.
- Let $G = (V, E)$ be a graph. A *chord* of a cycle of G is an edge that joins two nonconsecutive vertices of the cycle. A chord is thus an edge that joins two vertices of the cycle, but which is not itself an edge of the cycle. A cycle of length 3 cannot have any chords. A graph is *chordal*, provided that each cycle of length greater than 3 has a chord. A chordal graph has no chordless cycles. An induced subgraph of a chordal graph is also a chordal graph.



Independence number and clique number VII

- A special class of chordal graphs arises by considering intervals on aline. A close interval on the real line is denoted by

$$[a, b] = \{x : a \leq x \leq b\}.$$

Let

$$I_1 = [a_1, b_1], I_2 = [a_2, b_2], \ldots, I_n = [a_n, b_n]$$

be a family of closed intervals. Let G be the graph whose set of vertices is $\{I_1, I_2, \ldots, I_n\}$, where two intervals I_i and I_j are adjacent if and only if $I_i \cap I_j \neq \emptyset$. Such a graph G is called a *graph of intervals*, and any graph isomorphic to a graph of intervals is called an *interval graph*, Thus, the vertices of an interval graph can be thought of as intervals with two vertices adjacent if and only if the intervals have at least one point in common.

Theorem

Every interval graph is a chordal graph.

Independence number and clique number VIII

Lemma

Let $G = (V, E)$ be a connected graph and let U be an articulation set of G such that the subgraph G_U induced by U is a complete graph. Let the connected components of the induced subgraph G_{V-U} be $G_1 = (U_1, E_1), \ldots, G_t = (U_t, E_t)$. Assume that the induced graphs $G_{U_i \cup U}$ satisfy

$$\chi(G_{U_i \cup U}) = \omega(G_{U_i \cup U}) \quad (1, 2, \ldots, t).$$

Then

$$\chi(G) = \omega(G).$$



Independence number and clique number IX

- An articulation set U is a *minimal articulation set*, provided that, for all subsets $W \subset U$ with $W \neq U$, W is not an articulation set.

Theorem

Let $G = (V, E)$ be a connected chordal graph and let U be a minimal articulation set of G. Then the subgraphs G_U induced by U is a complete graph.

Theorem

Every chordal graph is perfect.

Corollary

Every interval graph is a perfect graph.



Independence number and clique number X

- A considerable amount of effort has been expended in attempts to characterize perfect graphs. These efforts have been largely directed toward resolving the following *conjecture* of Berge.

  A graph G is perfect if and only if neither G nor its complementary graph $\overline{G}$ has an induced subgraph equal to a cycle of odd length greater than three without any chords.



Connectivity I

- Let $G = (V, E)$ be a graph of order n. If G is a complete graph K_n, then we define its vertex-connectivity to be

$$\kappa(K_n) = n - 1.$$

Otherwise, we define the *vertex-connectivity* of G to be

$$\kappa(G) = \min\{|U| : G_{V-U} \text{ is disconnected}\},$$

the smallest number of vertices whose removal leaves a disconnected graph. Equivalently, the connectivity of a noncomplete graph equals the smallest size of an articulation set. A noncomplete graph has a pair of nonadjacent vertices a and b. Removing all vertices different from a and b leaves a disconnected graph, and hence $\kappa(G) \leq n - 2$ if G is a noncomplete graph of order n.



Connectivity II

Theorem

Let G be a graph of order n. Then

$$0 \leq \kappa(G) \leq n-1,$$

with equality on the left if and only if G is disconnected and with equality on the right if and only if G is a complete graph.

- The *edge-connectivity* of a graph G is defined to be the minimum number of edges whose removal disconnects G and is denoted by $\lambda(G)$. A connected graph G has edge-connectivity equal to 1 if and only if it has a bridge. The edge-connectivity of a complete graph K_n satisfies $\lambda(K_n) = n-1$. The edge-connectivity of a graph G satisfies $\lambda(G) \leq \delta(G)$, where $\delta(G)$ denotes the smallest degree of a vertex of G.



Connectivity III

Theorem

For each graph G, we have

$$\kappa(G) \leq \lambda(G) \leq \delta(G).$$

- Let G be a graph. Then G is connected if and only if its vertex-connectivity satisfies $\kappa(G) \geq 1$. if k is an integer and $\kappa(G) \geq k$, then G is called *k-connected*. We define an *articulation vertex* of a graph G to be a vertex a whose removal disconnects G, that is, a vertex such that $\{a\}$ is an articulation set.



Connectivity IV

Theorem

Let G be a graph of order $n \geq 3$. Then the following three assertions are equivalent:

1. G is 2-connected.
2. G is connected and does not have an articulation vertex.
3. For each triple of vertices a, b, c, there is a path joining a and b that does not contain c.

- Let $G = (V, E)$ be a connected graph of order $n \geq 2$. A *block* of G is a maximal induced subgraph of G that is connected and has no articulation vertex. More precisely, let U be a subset of the vertices of G. Then the induced subgraph G_U is a block of G, provided that G_U is connected and has no articulation vertex, and for all subsets W of the vertices of G with $U \subset W$ and $U \neq W$, either the induced subgraph G_W is not connected or it has an articulation vertex. It follows from Theorem 13.5.3 that the blocks of G are either the complete graph K_2 or are 2-connected.



Connectivity V

Theorem

Let $G = (V, E)$ be a connected graph of order $n \geq 2$, and let

$$G_{U_1} = (U_1, E_1), G_{U_2} = (U_2, E_2), \ldots, G_{U_r} = (U_r, E_r)$$

be the blocks of G. Then $E_1, E_2, \ldots, E_r$ is a partition of the set E of edges of G, and each pair of blocks has, at most, one vertex in common.

Theorem

Let $G = (V, E)$ be a graph of order $n \geq 3$. Then G is 2-connected if and only if, for each pair a, b of distinct vertices, there is a cycle containing both a and b.



Connectivity VI

Corollary

Let G be a graph with at least 3 vertices. Then G is 2-connected if and only if, for each pair a, b of distinct vertices, there are two paths joining a and b whose only common vertices are a and b.

Theorem

Let k be a positive integer and G be a graph of order $n \geq k + 1$. Then G is k-connected if and only if, for each pair a, b of distinct vertices, there are k paths joining a and b such that each path has only the vertices are a and b in common.

